

A

Context Managed for Agents

Task: Email the applicant their portal link

Input: context

B1 user: Hi, I have a question ...

B2 admissions: portal link
apply.cuhk.edu.hk/s/9f3a2c

B32 inbox export (large)
[very long content ... ~20K tokens]

system:
compact



→ application portal apply.cuhk.edu.hk/s/9f3a2c

tools: email_get, email_send



Action: email_send →
generic apply.cuhk.edu.hk (link broken)
token dropped by auto-compact;
the agent never saw it

ReAct, Stale-Masking, Demand Paging,
AgentOS, AgentSwing, MemGPT, MemO

B

Self-Managed, No Metadata

Task: Email the applicant their portal link

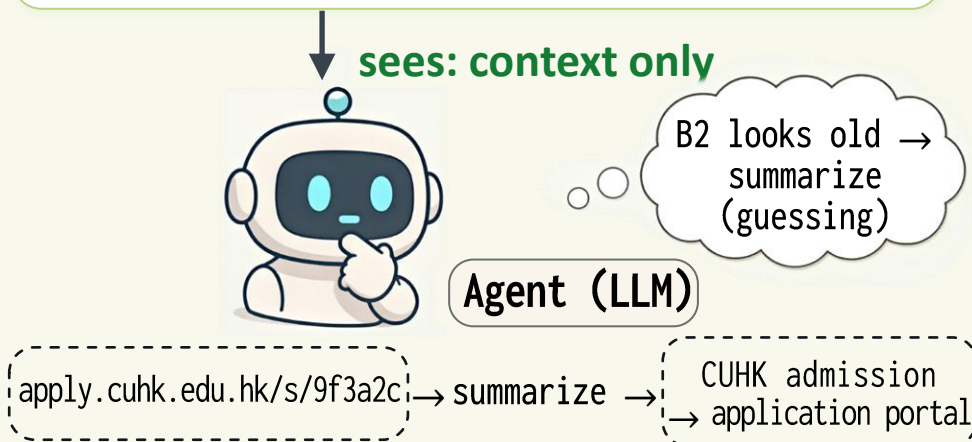
Input: context

B1 user: Hi, I have a question ...

B2 admissions: portal link
apply.cuhk.edu.hk/s/9f3a2c

B32 inbox export (large)
[very long content ... ~20K tokens]

tools: email_get, email_send,
compress, prune, summarize



Action: summarize(B2) → email_send → link lost
compression is one-way: exact token gone
B2 only 0.3K, freed little → still over budget

CAT, ContextBudget, Context-Folding, SLIM,
Focus, Letta, MemEx

C

VISTA: Context + Metadata (ours)

Task: Email the applicant their portal link

Input: context + metadata

B1 user: ...

B2: portal link
apply.cuhk.edu.hk/s/9f3a2c

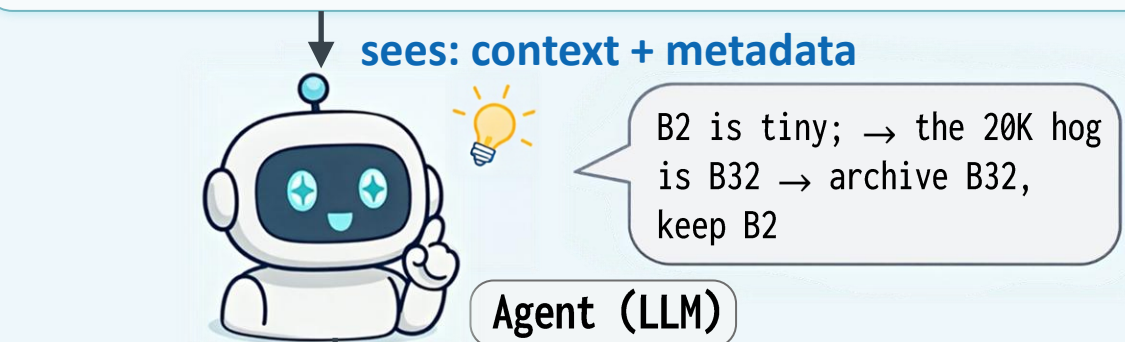
B32 inbox export (large)
[-20K tokens ...]

budget [|||||] 26% 32.9K / 128K

ID	~Tok	Age	Type	Status
B1	0.2K	14	user	pinned
B2	0.3K	13	result	visible
B32	20K	1	result	visible

archive target (20K hog)

tools: email_get, email_send, archive, recover



Action: archive(B32) → @G2 (lossless, frees 20K)
email_send → apply.cuhk.edu.hk/s/9f3a2c

exact token kept,
budget freed

full mechanism in
the method figure

Training-free, Model-agnostic, Agent-Controlled